RAJALAKSHMI ENGINEERING COLLEGE

SUBJECT:	OBJECT ORIENTED PROGRAMMING
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BONAFIDE CERTIFICATE

Certified that this project **"Expenditure Calculator"** is the bonafide work of **"MONICA N, MITHRAN VT"** who carried out the project work under my supervision.

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This mini project report is submitted for the viva voce examination to be held on

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

The *Expenditure Calculator* is a mini project developed to simplify the process of managing personal finances. It allows users to record daily expenses, categorize them, and view total and category-wise expenditure details. The project is built using **Java** for the frontend and **MySQL** for the backend, ensuring a smooth and secure data management system. By providing an easy-to-use interface and accurate tracking features, the project helps users maintain better financial discipline and awareness. It also highlights the practical use of database connectivity through **JDBC**, demonstrating the integration of frontend and backend technologies in real-world applications. By automating the expense tracking process, the project reduces manual errors and helps users gain better control over their financial habits.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The Expenditure Calculator is a simple financial management tool designed to help users track and manage their daily expenses efficiently. It records expenditures, categorizes spending, and provides total and category-wise analysis. The system aims to improve financial awareness and budgeting among users using a user-friendly interface.

1.2 SCOPE OF THE WORK

The *Expenditure Calculator* aims to provide users with an efficient and user-friendly way to record, manage, and analyze their daily, weekly, or monthly expenses. The project focuses on helping users maintain better financial awareness and budgeting habits by keeping track of all their spending activities

1.3 PROBLEM STATEMENT

Managing personal expenses manually can be time-consuming and inaccurate, often leading to poor financial planning. There is a need for a system that can simplify expense tracking and provide clear insights into spending habits.

1.4 AIM AND OBJECTIVES OF THE PROJECT

The aim of this project is to design and develop an Expenditure Calculator that helps users efficiently record, manage, and analyze their daily expenses. The system aims to promote better financial planning and awareness through an organizedand user-friendly interface.

CHAPTER 2

SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	:	Intel i5
Memory Size	:	8GB (Minimum)
HDD	:	1 TB (Minimum)

2.2 SOFTWARE SPECIFICATIONS

Operating System	:	WINDOWS 10
Front – End	:	Java Swing
Back - End	:	MySql
Language	:	Java,SQL

CHAPTER 3

MODULE DESCRIPTION

This application consists of two modules. When the program runs, it will ask for a confirmation to the login window. The person who interacts can login as an Administrator or as a User. The description of the modules are as follows:

1. **Admin login**

   When the person who interacts tries to login as Admin then he needs to login with his username and password. The administrator only has the power to change and manipulate the data in the database.

2. **User login**

   When the person tries to login as a user then he/she will be prompted to enter the number of symptoms and the final result will be printed in the form of table or in any form of a graph.

CHAPTER 4

SAMPLE CODING

package dashboard:

```
import expensetracker.DatabaseManager;

import expensetracker.UserSession;

import javax.swing.*;

import java.awt.*;

import java.sql.Connection;

import java.sql.PreparedStatement;

import java.sql.SQLException;

public class TransactionForm extends JPanel {

    private final Color primaryGreen = new Color(76, 175, 80);

    private final Color lightGray = new Color(240, 240, 240);

    private final Color textColor = new Color(51, 51, 51);

    private JTextField amountField;

    private JComboBox<String> typeComboBox;

    private JComboBox<String> categoryComboBox;

    private JTextArea descriptionArea;

    private JTextField dateField;
```

```
private JButton saveButton;

private DatabaseManager dbManager = new DatabaseManager();

// Reference to the parent Dashboard frame

private Dashboard parentFrame;

public TransactionForm(Dashboard parent) {

    this.parentFrame = parent; // Store the reference

    setBackground(lightGray);

    setLayout(new GridBagLayout());

    setBorder(BorderFactory.createEmptyBorder(30, 50, 30, 50));

    GridBagConstraints gbc = new GridBagConstraints();

    gbc.insets = new Insets(10, 10, 10, 10);

    gbc.fill = GridBagConstraints.HORIZONTAL;

    JLabel titleLabel = new JLabel("Record New Transaction", SwingConstants.LEFT);

    titleLabel.setFont(new Font("Arial", Font.BOLD, 28));

    titleLabel.setForeground(textColor);

    gbc.gridx = 0;

    gbc.gridy = 0;

    gbc.gridwidth = 2;

    gbc.anchor = GridBagConstraints.WEST;
```

```
add(titleLabel, gbc);

gbc.gridy = 1;

gbc.insets = new Insets(0, 0, 20, 0);

add(Box.createVerticalStrut(20), gbc);

gbc.insets = new Insets(10, 10, 10, 10);

gbc.gridwidth = 1;

gbc.gridx = 0;

gbc.gridy = 2;

add(createLabel("Transaction Type:"), gbc);

typeComboBox = new JComboBox<>(new String[]{"Expense", "Income"});

typeComboBox.setFont(new Font("Arial", Font.PLAIN, 14));

gbc.gridx = 1;

add(typeComboBox, gbc);

gbc.gridx = 0;

gbc.gridy = 3;

add(createLabel("Amount ($):"), gbc);

amountField = new JTextField(15);

amountField.setFont(new Font("Arial", Font.PLAIN, 14));

gbc.gridx = 1;
```

```
        add(amountField, gbc);

        gbc.gridx = 0;

        gbc.gridy = 4;

        add(createLabel("Category:"), gbc);

        String[] categories = {"Groceries", "Salary", "Rent", "Utilities", "Entertainment",
"Other"};

        categoryComboBox = new JComboBox<>(categories);

        categoryComboBox.setFont(new Font("Arial", Font.PLAIN, 14));

        gbc.gridx = 1;

        add(categoryComboBox, gbc);

        gbc.gridx = 0;

        gbc.gridy = 5;

        add(createLabel("Date (YYYY-MM-DD):"), gbc);

        dateField = new JTextField(15);

        dateField.setFont(new Font("Arial", Font.PLAIN, 14));

        gbc.gridx = 1;

        add(dateField, gbc);

        // --- Description ---

        gbc.gridx = 0;

        gbc.gridy = 6;
```

```
add(createLabel("Description:"), gbc);

descriptionArea = new JTextArea(3, 15);

descriptionArea.setFont(new Font("Arial", Font.PLAIN, 14));

descriptionArea.setBorder(BorderFactory.createLineBorder(Color.LIGHT_GRAY));

JScrollPane scrollPane = new JScrollPane(descriptionArea);

gbc.gridx = 1;

gbc.weighty = 0.5;

add(scrollPane, gbc);

// --- Save Button ---

gbc.gridy = 7;

gbc.gridx = 0;

gbc.gridwidth = 2;

gbc.weighty = 0.0;

gbc.insets = new Insets(20, 10, 10, 10);

saveButton = new JButton("SAVE TRANSACTION");

saveButton.setFont(new Font("Arial", Font.BOLD, 16));

saveButton.setBackground(primaryGreen);

saveButton.setForeground(Color.WHITE);

saveButton.setFocusPainted(false);
```

```
        saveButton.setBorder(BorderFactory.createEmptyBorder(12, 0, 12, 0));

        // Link action to the save method

        saveButton.addActionListener(e -> saveTransaction());

        add(saveButton, gbc);

    }

    private JLabel createLabel(String text) {

        JLabel label = new JLabel(text + " ");

        label.setFont(new Font("Arial", Font.BOLD, 14));

        label.setForeground(textColor);

        label.setHorizontalAlignment(SwingConstants.RIGHT);

        return label;

    }
```

Sample 1

Sample 1 depicts the display code, that gets the data from the database i.e. being stored there and represented on users demand with the layout and measurements.

```
package expensetracker;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.PreparedStatement;
```

```
import java.sql.ResultSet;

import java.sql.SQLException;

public class DatabaseManager {

    private static final String DB_URL =
"jdbc:mysql://localhost:3306/expense_income_db";

    private static final String USER = "root";

    private static final String PASS = "Solamata@18";

     * Establishes a connection to the MySQL database.

     * @return A live Connection object, or null if the connection fails due to bad
credentials/path.

    public static Connection getConnection() {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            return DriverManager.getConnection(DB_URL, USER, PASS);

        } catch (ClassNotFoundException e) {

            System.err.println("MySQL JDBC Driver not found. Please ensure the JAR
is correctly added.");

            return null;

        } catch (SQLException e) {
```

```
        System.err.println("Database connection failed. Check URL, USER, and
PASS variables in DatabaseManager.java.");

        e.printStackTrace();

        return null;

    * Authenticates a user against the 'users' table.

    * @return The user_id (int) on success, or -1 on failure.

    public int authenticateUser(String username, String password) {

        String sql = "SELECT user_id FROM users WHERE username = ? AND
password = ?";

        try (Connection conn = getConnection();

             PreparedStatement pstmt = conn.prepareStatement(sql)) {

            if (conn == null) return -1;

            pstmt.setString(1, username);

            pstmt.setString(2, password);

            try (ResultSet rs = pstmt.executeQuery()) {

                if (rs.next()) {

                    return rs.getInt("user_id"); // Return the user's unique ID

                }

                return -1;
```

```
        }
        } catch (SQLException e) {
            System.err.println("Authentication query failed.");
            e.printStackTrace();
            return -1;
        }
    }
}
```

Sample 2

Sample 2 depicts the calculation of expenditures where it displays the expenses and its details and stored in the database.

CHAPTER 5

SCREEN SHOTS

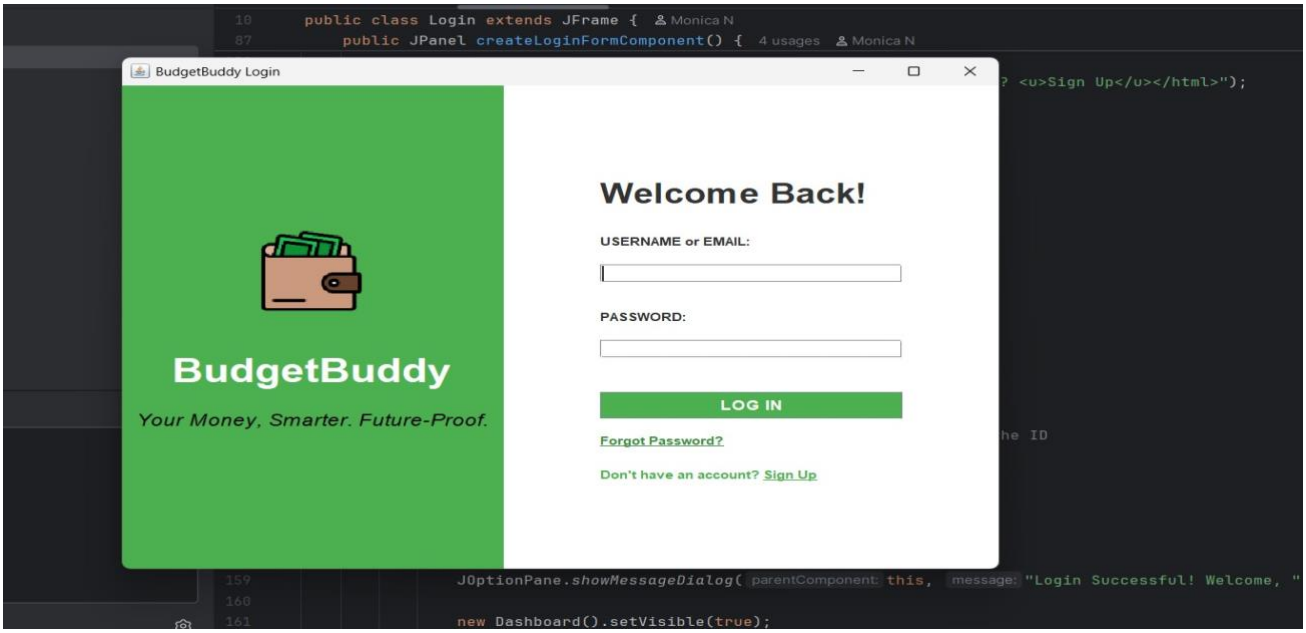

Fig 5.1 Login page

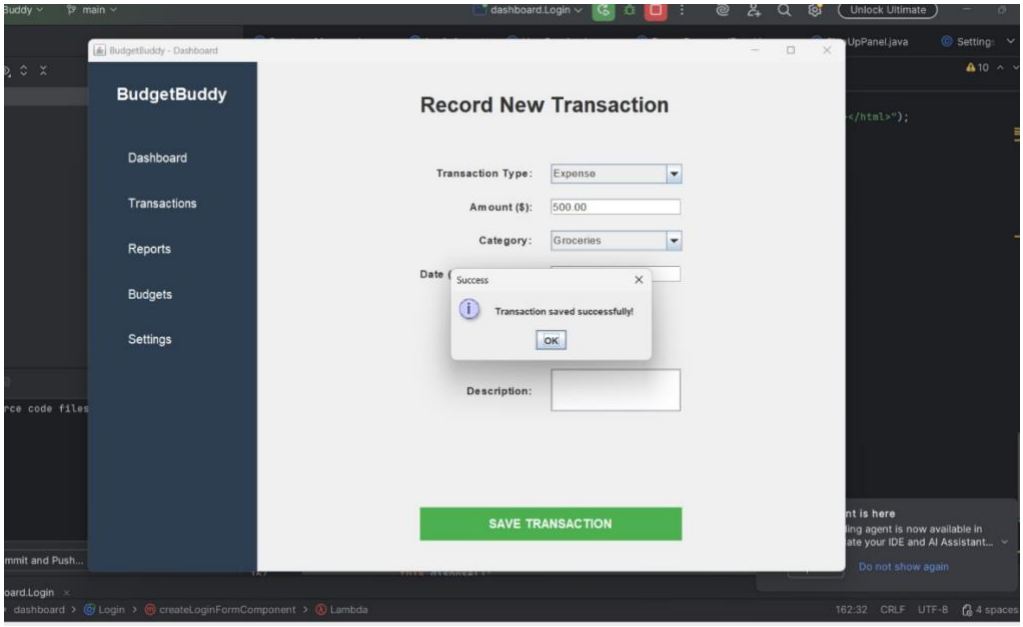

Fig 5.2 Expense details

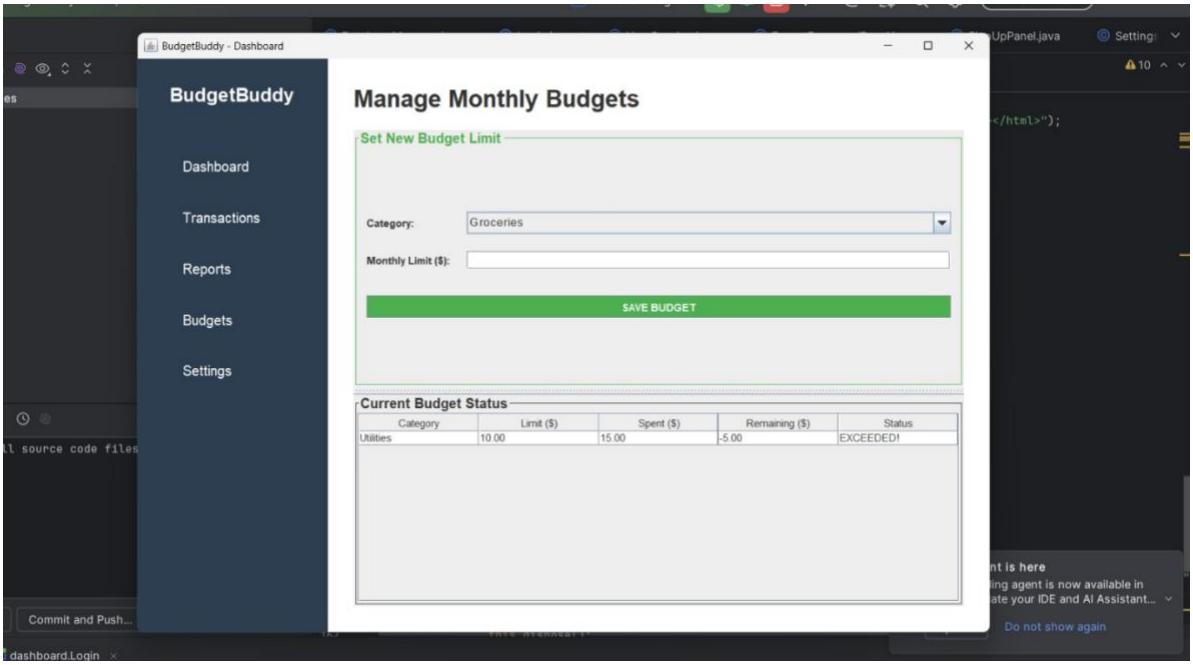

Fig 5.3 Expense Log

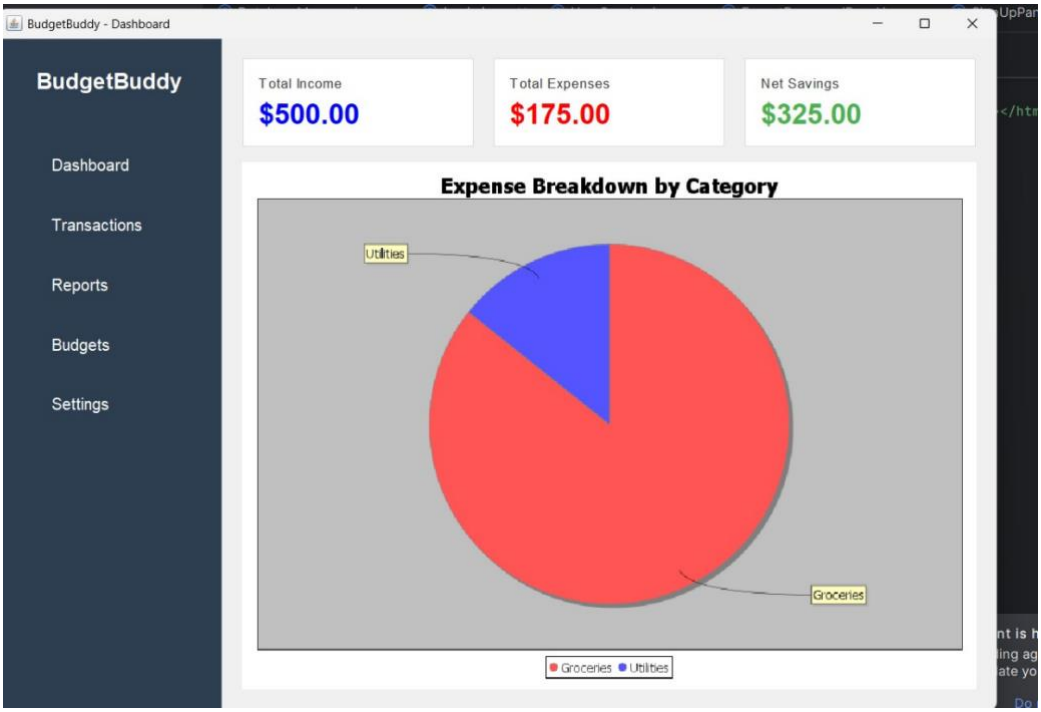

Fig 5.4 Analysis of Expenses

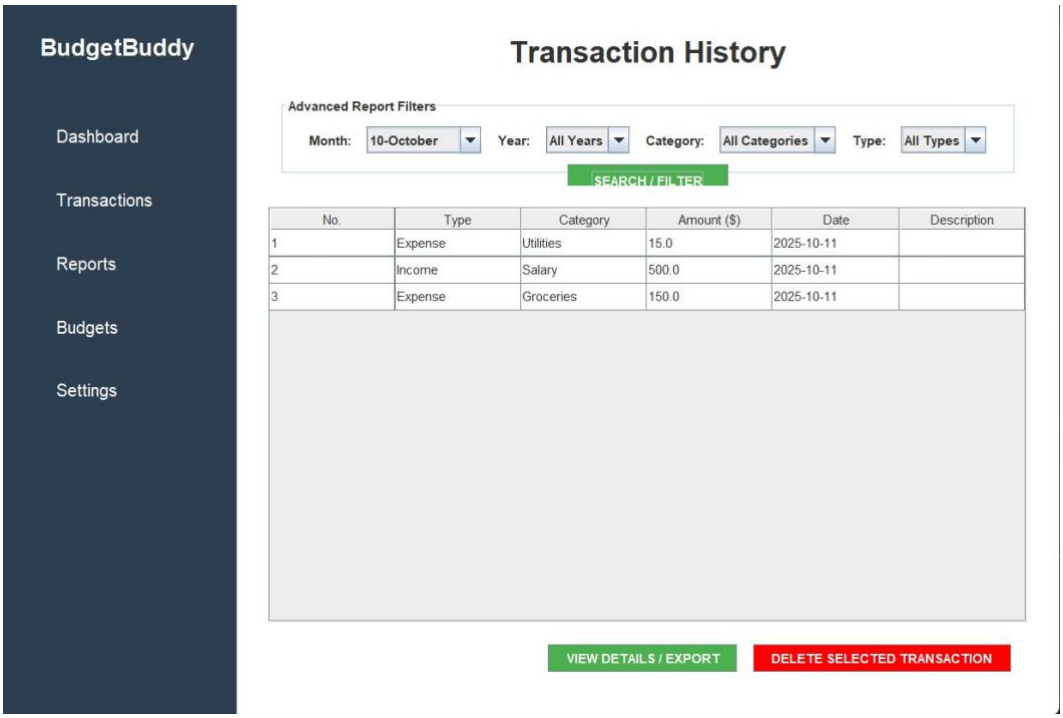

Fig 5.5 EXPENSE DATABASE

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion:

The *Expenditure Calculator* project successfully provides a simple and effective solution for managing daily expenses. It allows users to record, categorize, and analyze their spending, helping them gain better control over their finances. The system fulfills its objective of promoting financial awareness through an easy-to-use interface and accurate expense tracking.

Future Enhancement:

- Integration with mobile applications or cloud storage for data access anywhere.
- Addition of charts and graphs for visual expense analysis.
- Implementation of user authentication for personalized profiles.
- Automatic budget suggestions based on spending patterns.
- Option to export expense reports in PDF or Excel format.

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